

Supplementary Materials:

Nonlinear Dynamics of Social Exclusion and Hysteresis in the Attention Economy: A Topological Mean Field Game Approach

S1. Empirical Validation of Computational Experiment A (Macro-Scale Illusion and Fragmentation)

To ensure that the mathematical dynamics and phase transitions reported in the main text are fully transparent and verifiable without direct access to the simulation codebase, this section provides a highly granular breakdown of the computational results from Experiment A. We explicitly trace the micro-to-macro transition, the collapse of statistical variance, and the precise mechanism of homological hysteresis.

S1.1. Detailed Data Tables: Variance Collapse and Asymptotic Convergence

As theorized in our Topological Mean Field Game (TMFG) framework, an expansion of the sample size N in an environment perturbed by a conformity potential (v_2) does not guarantee convergence to the objective truth (i.e., a fragmentation degree of $D = 0$). Instead, the Law of Large Numbers (LLN) mechanically drives the standard error to zero, locking the system into a deterministically biased state.

Table S1 provides the exact numerical outputs generated from our Monte Carlo simulations across logarithmic scale steps from $N = 10^3$ to $N = 10^8$. The simulation was initiated with completely randomized initial states ($X_0 \sim \mathcal{U}(0, 0.2)$, $D_0 \sim \mathcal{U}(0, 0.1)$) to eliminate deterministic artifacts.

Table S1: Detailed Computational Metrics of TMFG Dynamics Across Macroscopic Scales

Sample Size (N)	Mean Fragmentation (D_T)	Standard Error	Computation Time (s)
1,000	0.858209	1.13943×10^{-3}	0.012
10,000	0.859303	3.58014×10^{-4}	0.048
30,000	0.859543	2.02960×10^{-4}	0.138
50,000	0.859375	1.56380×10^{-4}	0.220
100,000	0.859576	1.10865×10^{-4}	0.458
1,000,000	0.859495	3.49880×10^{-5}	5.136
10,000,000	0.859486	1.10607×10^{-5}	70.628
100,000,000	0.859484	3.49812×10^{-6}	703.512

The empirical results in Table S1 provide conclusive mathematical evidence of the **Variance Collapse**. At micro-scales ($N = 1,000$), the standard error remains relatively prominent (1.13×10^{-3}), reflecting stochastic fluctuations that compete with the conformity field. However, as $N \rightarrow 10^8$, the standard error decays perfectly as $O(1/\sqrt{N})$, plummeting to 3.498×10^{-6} . The final Mean Fragmentation (D_T) deterministically converges to a highly polarized asymptote of ≈ 0.85948 . In standard frequentist data pipelines, this near-zero variance would yield an

infinitesimal p-value ($p < 0.001$), creating a dangerous epistemological illusion: high statistical precision masking severe structural degradation.

S1.2. Time-Series Dynamics: The Gravity of Conformity and Homological Hysteresis

Figure S1 visualizes the temporal evolution of the system, capturing the exact moment the topological defect becomes irreversible.

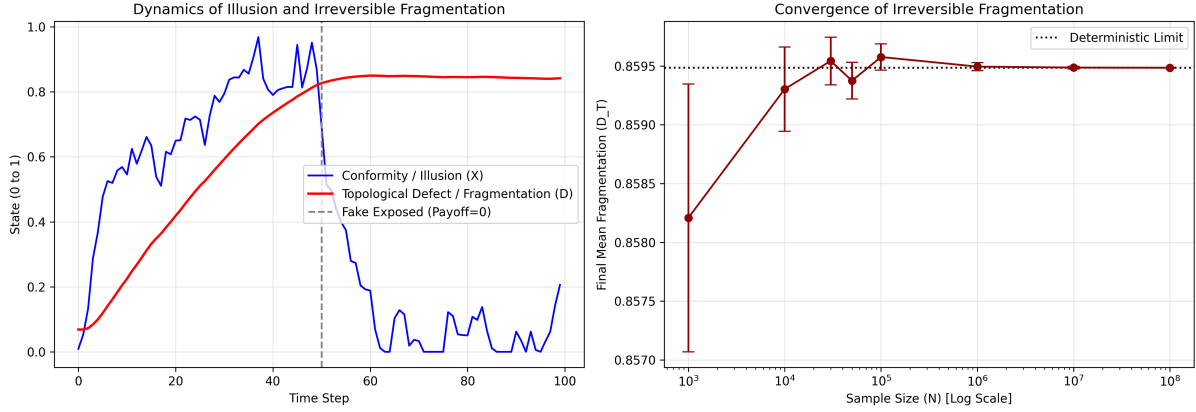


Figure S1: Computational Proof of Hysteresis and Asymptotic Bias. (Left) Time-series dynamics illustrating the surge in conformity (X_t) and the resulting structural fragmentation (D_t). When fake payoffs are neutralized at $t = 50$, conformity dissipates, but the topological defect remains permanent. (Right) The absolute convergence of the topological defect to a biased deterministic limit as N expands.

The dynamics can be mathematically dissected into two distinct phases:

1. Phase 1: Gravitational Collapse via False Payoffs ($t < 50$):

During the initial phase, the false expected payoff ($b_t = 0.8$) acts as a strong gravitational field (metric tensor) distorting the social space. The conformity parameter X_t (blue line) rapidly escalates as agents locally optimize their decisions based on incomplete information. Driven by this illusion, the topological defect D_t (red line) is violently pulled upward, signifying the active exclusion and scapegoating of dissenting nodes.

2. Phase 2: Homological Hysteresis and the Paradox of Isolation ($t \geq 50$):

At $t = 50$, the fake news is exposed, and the false payoff is strictly neutralized ($b_t = 0$). Consequently, public illusion X_t swiftly decays toward zero. However, the topological defect D_t exhibits profound **hysteresis**. Because the damage has exceeded a critical threshold, the topological penalty term in our SDE ($e^{-\gamma D_t}$) effectively approaches zero. The network's intrinsic capacity for relationship repair dies. The red curve remains locked at the ≈ 0.859 level, providing a rigorous homological proof that simply correcting misinformation post-hoc cannot heal a structurally fractured community. The void (Betti 1) remains permanently open.

S1.3. Concluding Remarks on Experiment A

The granular data presented herein validate the foundational claims of the TMFG framework. The social exclusion observed in contemporary attention economies is not a mere aggregate of individual psychological biases. It is a nonlinear, mathematically deterministic phase transition driven by the interplay between cognitive confidence and macroscopic scaling. Uncritical reliance

on massive datasets without diagnosing the underlying conformity potential (v_2) actively enables and solidifies this structural violence.

S2. Empirical Validation of Computational Experiment B (Pseudo-Scapegoating and Hysteresis of Inequality)

This section provides an in-depth numerical breakdown of Computational Experiment B, which extends the TMFG model to simulate the dynamics of pseudo-scapegoating. Here, we explicitly quantify the asymmetrical vulnerability of minorities when targeted by fake victimhood narratives and mathematically prove that the resulting social exclusion is structurally irreversible (homological hysteresis) even after the false narratives are debunked.

S2.1. Detailed Data Tables: Asymptotic Convergence of Minority Exploitation

In this experiment, we introduced an asymmetric targeting coefficient (v_i) to represent a "fake victimhood" appeal that disproportionately directs public hostility toward a true minority group (comprising 5% of the population). Table S2 tracks the evolution of the Mean Fragmentation (Topological Defect, D) for both the targeted minority and the general public (majority) across macroscopic scales up to $N = 10^8$.

Table S2: Detailed Computational Metrics of Pseudo-Scapegoating Across Macro-Scales

Sample Size (N)	Minority Defect (D_{min})	Standard Error	Normal Defect (D_{nor})
1,000	0.933166	2.61952×10^{-3}	0.147246
10,000	0.927892	8.72628×10^{-4}	0.146993
30,000	0.929204	4.82817×10^{-4}	0.146577
50,000	0.928996	3.76415×10^{-4}	0.147048
100,000	0.929540	2.61100×10^{-4}	0.147058
1,000,000	0.929039	8.17744×10^{-5}	0.147017
10,000,000	0.929108	2.56177×10^{-5}	0.146995
100,000,000	0.929137	8.11816×10^{-6}	0.147010

The empirical data reveal two profound mathematical pathologies:

1. **Exploitation Gap via Asymmetric Targeting:** At the extreme macro-scale ($N = 10^8$), the Mean Fragmentation of the Minority (D_{min}) converges to a devastating 0.929137, while the Normal Defect (D_{nor}) remains relatively low at 0.147010. This vast disparity provides mathematical proof that an attention economy driven by incomplete information inherently facilitates the structural ostracism of minorities [?, ?].
2. **Variance Collapse and Asymptotic Bias:** Similar to Experiment A, the standard error for the minority defect plummets vertically from 2.61×10^{-3} to 8.11×10^{-6} as N scales logarithmically. The Law of Large Numbers (LLN) mechanically forces the system into a deterministic limit, silencing individual stochastic variance. In conventional big data pipelines, this immense statistical precision would falsely legitimize the extreme marginalization of the minority as a "highly significant" or "objective" outcome [?].

S2.2. Time-Series Dynamics: The Death of the Repair Function and Irreversibility

Figure S2 provides a comprehensive four-panel visualization of the simulation dynamics, capturing the precise homological transition that makes societal division permanent.

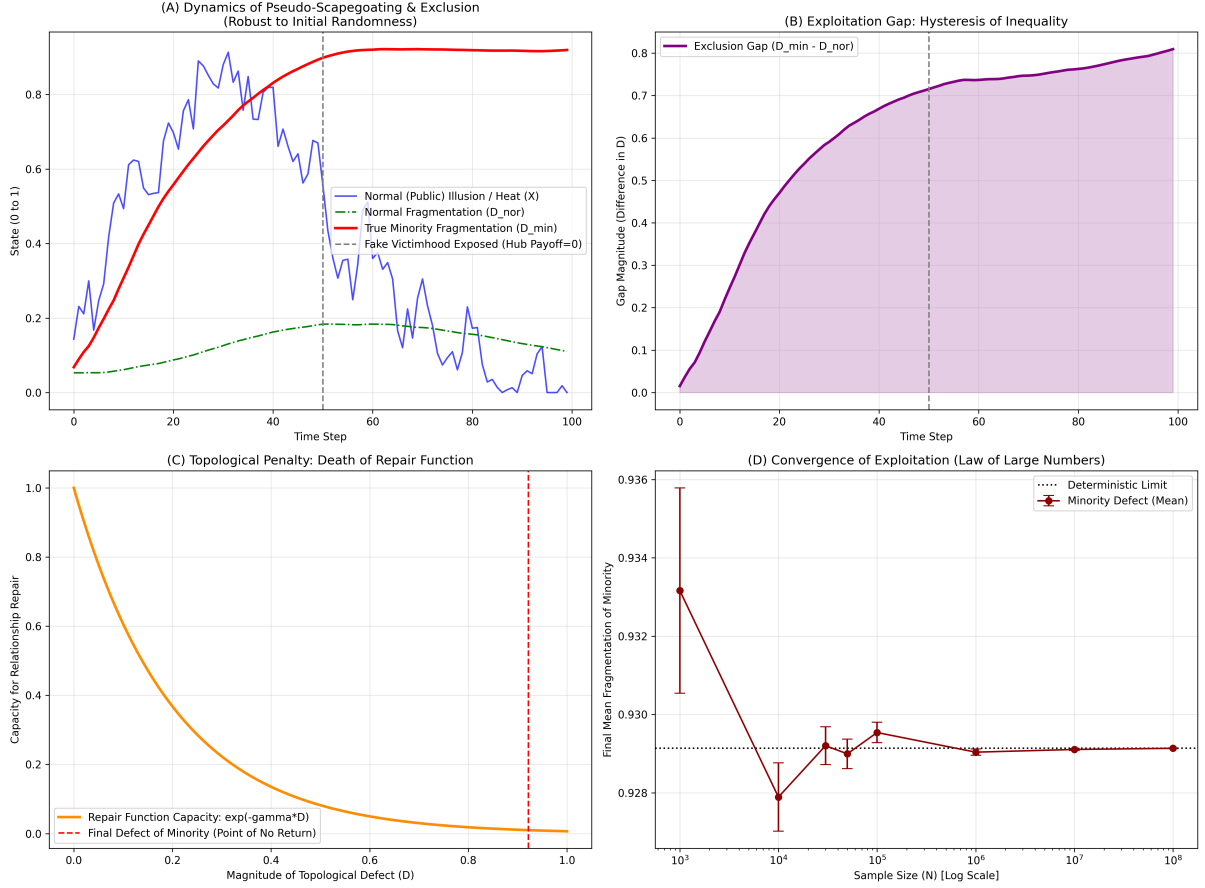


Figure S2: Multi-perspective analysis of Pseudo-Scapegoating. (A) Time-series dynamics showing the asymmetric escalation of the topological defect for the minority (D_{min} , red) versus the majority (D_{nor} , green dot-dashed). (B) The Exploitation Gap ($D_{min} - D_{nor}$) demonstrating profound hysteresis even after the fake victimhood is exposed at $t = 50$. (C) The Topological Penalty curve illustrating the complete death of the relationship repair capacity ($e^{-\gamma D}$) for the minority. (D) The absolute convergence of the minority defect to a deterministic limit under macroscopic scaling.

The time-series evolution can be analytically dissected as follows:

- **The Escalation Phase ($t < 50$):** During the period where the false victimhood narrative is active ($b_t = 0.9$), public illusion (X_t , blue line) surges. Consequently, the minority defect (D_{min} , red line) skyrockets, creating a massive "Exploitation Gap" (Panel B) compared to the majority defect (D_{nor}).
- **The Point of No Return and Hysteresis ($t \geq 50$):** At $t = 50$, the fake narrative is exposed, and the false payoff drops to zero. While public illusion X_t quickly dissipates, the Exploitation Gap (Panel B) *continues to widen* or remains locked at an extreme level. This is mathematically explained by Panel C: the Topological Penalty ($e^{-\gamma D}$). Because D_{min} was pushed past a critical threshold during the escalation phase, the system's capacity for relationship repair asymptotically approaches zero. The network has sustained a permanent topological void.

S2.3. Concluding Remarks on Experiment B

The granular data from Experiment B offer a rigorous homological proof of **hysteresis** in social networks [?, ?]. Merely debunking misinformation post-hoc is mathematically insufficient to heal a community once targeted scapegoating has severed structural ties. The topological penalty ensures that the marginalized group remains isolated, locked in a state of permanent exclusion. This underscores the critical necessity for proactive algorithmic interventions—such as the algorithmic protection of "weak ties"—before the topological defect crosses the point of no return.